Vegard Pettersen

Portfolio | GitHub | vegardbp@gmail.com | Norway

PROFILE

- BE in Automation & Intelligent Systems, NTNU 2026, average grade **4.3/5**, A grades in Control Theory, DSP, and Intelligent Systems.
- Built a hardware-accurate real-time simulator of a cyber-physical system (Sphero RVR), including a physics-based control system and monocular SLAM.
- GPU-accelerated optimization: applied a genetic algorithm to PID controller tuning and high-dimensional search problems (HLSL/GLSL compute).
- Strong C++/C# background building real-time systems and large solo codebases, including custom physics engines and networked simulations.

TECHNICAL SKILLS

Languages: C++, C, Python, C#, HLSL, GLSL
Engines & Tools: Git, VCPkg, CLion, Visual Studio, Linux, Unity, Blender
Specialties: Control Systems, Simulation, Physics, Optimization, Computer Vision (OpenCV), Compute Shaders, Multiplayer Networking

EDUCATION

Bachelor of Engineering, Automation *NTNU, 2023–2026*
Average Grade: **4.3 / 5.0** | A grades in: Control Theory, Intelligent Systems, DSP, Mathematics, Physics, and OOP.

SELECTED PROJECTS

RC Car Hardware-in-the-Loop Simulator | *Group Project* *2025*
Tech: C++, threepp, Bullet Physics, Python, Git

- Hardware-accurate real-time simulation of a Sphero RVR, indistinguishable from the physical robot to any external control system.
- Responsible for the full simulator and control system: vehicle dynamics via Bullet Physics, first-person 3D rendering via threepp, and live video streamed over UDP-IP.
- Implemented monocular SLAM without IMU data and object recognition (OpenCV) in Python, for both simulated and real-world objects.
- Graded **B**; simulation accuracy and SLAM implementation were noted as strengths.

Compute Shader Genetic Algorithm *Exam Project*
Tech: Unity, C#, HLSL, Git

- GPU-accelerated genetic algorithm solving the Rastrigin function in high-dimensional search spaces.
- Same architecture applied to 3-dimensional PID controller tuning against specified desired responses, solving in milliseconds.
- Graded **A**; noted for algorithm design and documentation.

Dark Bros | *Solo Developer* *Releasing Q3 2026*
Tech: Unity, C#, HLSL, Netcode for GameObjects, Steamworks

- Solo-built and shipped a real-time networked multiplayer application on Steam, covering networking, shaders, 3D models, animations, and UI, over 1.5 years.
- Networking built on Steam P2P via Facepunch Steamworks, with a full lobby system and a modular attack system that abstracts away networking for most actions.
- Snapshot-based replay system for full post-match playback of state; GPU-instanced rendering with compute shader culling.

WORK EXPERIENCE

Staff | ***Buldrehallen, Norway*** *2025 (6 months)*

- Ran solo shifts of 5 to 8 hours with full responsibility for safety, security, and customer experience.
- Handled POS transactions, membership registrations, and food and beverage service.
- Gave safety briefings and technical instruction to new climbers.
- Reference available upon request.